

NECEDAH 5 WNW, WI, USA

WMO: 743580

Lat: 44.0604N

Lon: 90.1737W

Elev: 284

StdP: 97.95

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54903

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.4	-27.6	0.3	-23.3	-25.2	0.4	-20.9	8.7	-10.5	7.8	-9.0	1.9	N/A	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	31.2	22.9	29.7	21.8	28.2	21.0	24.3	29.4	23.3	27.9	22.3	26.5	3.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.1	27.0	21.7	17.0	25.9	20.8	16.0	25.1	74.6	29.5	70.6	28.1	66.9	26.6	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.5	5.7	-28.9	33.5	3.7	1.6	-31.6	34.7	-33.8	35.6	-35.8	36.5	-38.5	37.7		
			-28.1	26.0	3.3	1.1	-30.4	26.8	-32.4	27.5	-34.2	28.1	-36.6	29.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-8.9	-7.1	0.4	7.8	14.6	19.6	21.7	20.3	16.2	9.2	2.0	-5.4
	DBStd	11.92	7.00	6.73	6.99	5.36	4.90	3.26	3.10	2.96	4.48	5.06	5.56	6.27
	HDD10.0	2294	585	479	307	104	13	0	0	5	78	247	477	
	HDD18.3	4217	843	712	556	318	137	23	7	14	93	287	492	735
	CDD10.0	1420	0	0	11	37	156	287	362	319	191	53	5	0
	CDD18.3	301	0	0	1	1	21	61	111	75	28	3	0	0
	CDH23.3	2581	0	0	7	23	216	500	988	576	239	31	0	0
	CDH26.7	682	0	0	0	3	54	122	312	133	54	3	0	0
Wind	WSAvg	2.5	2.8	2.8	2.8	3.2	2.8	2.3	1.9	1.8	1.9	2.4	2.6	2.6
Precipitation	PrecAvg	818	25	20	40	75	99	129	111	106	83	58	40	30
	PrecMax	1071	60	45	85	154	236	263	330	224	221	140	112	67
	PrecMin	593	9	2	2	20	40	38	34	29	20	12	1	4
	PrecStd	122	12	11	22	34	41	68	73	48	48	32	26	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	13.1	24.2	26.4	31.2	32.0	34.4	31.7	30.7	26.9	19.8	10.3
		MCWB	4.2	9.4	15.9	15.2	20.4	23.5	24.4	23.2	22.6	17.0	12.3	8.2
	2%	DB	5.2	7.6	17.9	22.6	28.0	29.8	31.8	29.7	28.3	23.4	16.0	7.0
		MCWB	2.9	5.1	12.0	14.1	18.2	21.4	24.0	22.2	21.4	16.1	10.9	4.9
	5%	DB	2.7	4.6	14.3	19.9	25.7	27.9	29.9	28.1	26.1	20.7	13.0	4.8
		MCWB	1.1	2.1	9.2	11.6	17.1	20.1	22.4	21.5	19.6	15.2	8.6	3.1
	10%	DB	0.9	2.0	10.3	17.1	23.5	26.3	28.2	26.6	24.1	17.9	10.4	2.7
		MCWB	-0.1	0.0	6.2	9.9	16.3	19.4	21.6	20.5	18.7	13.8	6.9	1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	8.4	16.8	16.9	22.3	24.8	26.4	24.7	23.7	19.8	13.9	9.6
		MCDB	7.2	12.6	23.4	23.1	29.0	30.0	31.5	29.2	29.0	21.8	17.1	10.3
	2%	WB	2.9	5.5	14.3	14.7	20.5	23.0	24.9	23.7	22.3	17.5	11.4	5.5
		MCDB	4.9	8.1	17.4	21.0	26.1	28.2	29.9	28.0	26.8	20.2	16.0	6.2
	5%	WB	1.3	2.5	9.8	12.6	19.1	21.7	23.7	22.6	21.0	15.6	9.3	3.2
		MCDB	2.5	4.7	13.6	17.8	24.1	26.0	28.7	26.5	24.1	19.6	12.5	4.9
	10%	WB	0.0	0.4	6.7	10.3	17.6	20.4	22.6	21.6	19.8	13.6	7.0	1.5
		MCDB	1.1	2.3	10.7	15.9	22.2	24.7	26.8	25.4	22.8	17.1	9.7	2.8
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	10.4	10.5	11.1	10.5	11.1	10.9	11.5	10.2	8.6	7.6
		MCDBR	10.1	11.4	14.0	15.5	14.1	12.5	12.7	12.0	13.2	13.5	13.0	9.7
	5% WB	MCWBR	8.2	8.2	8.0	7.9	6.4	5.9	5.5	5.4	6.5	7.8	8.4	7.6
		MCDWR	9.2	10.9	11.7	13.3	11.8	10.6	10.8	10.2	10.5	11.9	12.4	8.3
Clear-Sky Solar Irradiance	taub		0.259	0.290	0.323	0.358	0.389	0.399	0.393	0.397	0.362	0.325	0.290	0.265
		taud	2.399	2.309	2.378	2.355	2.315	2.316	2.341	2.331	2.423	2.516	2.529	2.452
	Ebn at Noon		881	903	916	906	882	871	872	856	863	857	840	840
		Edh at Noon	74	97	105	117	126	126	122	119	100	80	66	64
All-Sky Solar Radiation	RadAvg		1.74	2.85	3.92	4.51	5.39	5.94	6.11	5.22	4.18	2.66	1.72	1.28
	RadStd		0.20	0.26	0.46	0.43	0.41	0.45	0.30	0.35	0.32	0.32	0.16	0.14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page